

5th order IIR typically looks like.

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + b_4 z^{-4} + b_5 z^{-5}}{1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + a_4 z^{-4} + a_5 z^{-5}}$$

5 zeros (numerator roots), 5 poles (denominator roots) - this is the starting point.

Cascade

What it is: factor the numerator and denominator each into smaller 1st & 2nd order polynomials, we pair them up & implement $H(z)$ as a series of connection of filter ~~functions~~ sections

$$H(z) = \prod_{k=1}^K \frac{b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2}}{1 + a_{1k} z^{-1} + a_{2k} z^{-2}}$$

Why for a 5th order filter (odd order), ~~that~~ we'll get two 2nd order + One 1st order section, each implemented as it's own Small Direct form II, wired o/p $\rightarrow$ i/p in a chain.

Advantages:

- 1) Each section only has 2 poles/zeros $\rightarrow$ much lower co-efficient sensitivity $\Rightarrow$ a rounding error in 1 section's co-efficient barely affects the others.

2. Easier to design/tune: you can adjust 1 section (eg. move a troubled pole pair) without redesigning the whole filter.
3. Complex-conjugate pole/zero pairs (common in real designs) fit naturally into 2nd order sections with real co-efficients.

Disadvantages

1. Signal scaling / dynamic range issues - since sections are in series, the signal amplitude can grow or shrink a lot as it passes through each stage, risking overflow or underflow in fixed-point H/W.
2. The order in which we arrange the section matters: different orderings / of poles & zeroes would give different internal signal levels and noise behavior, so this needs careful design choices.

Parallel:

What it is: parallel fraction expansion used to break $H(z)$ into a sum of small sections instead of series.

$$H(z) = C_0 + \frac{A_1}{1 + \alpha_{11}z^{-1}} + \frac{\gamma_{02} + \gamma_{12}z^{-1}}{1 + \alpha_{12}z^{-1} + \alpha_{22}z^{-2}} + \dots$$

Why: for your 5th order filter, this gives (one 1st order single pole, plus 2nd order terms (each holding one conjugate pole pair)) all computed independently and added together at the end; not chained in series.

Advantages 1) Each sections o/p error/noise doesn't propagate into other sections (~~they~~ independent, only combined at the final adder); generally better numerical behavior than cascade.

2. Sections can be computed in parallel (simultaneously) in H/w, which can reduce total computation time even though total work is similar.

3. No "ordering" design decision needed, unlike cascade, sections don't ~~interact~~ interact with each other.

Disadvantages:

- 1) Cannot directly place zeros anywhere: the zeroth location of the overall $H(z)$ come out as consequence of the partial fraction terms, not something ~~we~~ directly controlled the way it's done in cascade.
- 2) If the numerator order $\geq$ denominator order; we get extra direct transmission terms (C_0, C, z^{-1} , etc) adding more complexity.
- 3) Slightly more adders needed overall as in we sum all the sections eventually.